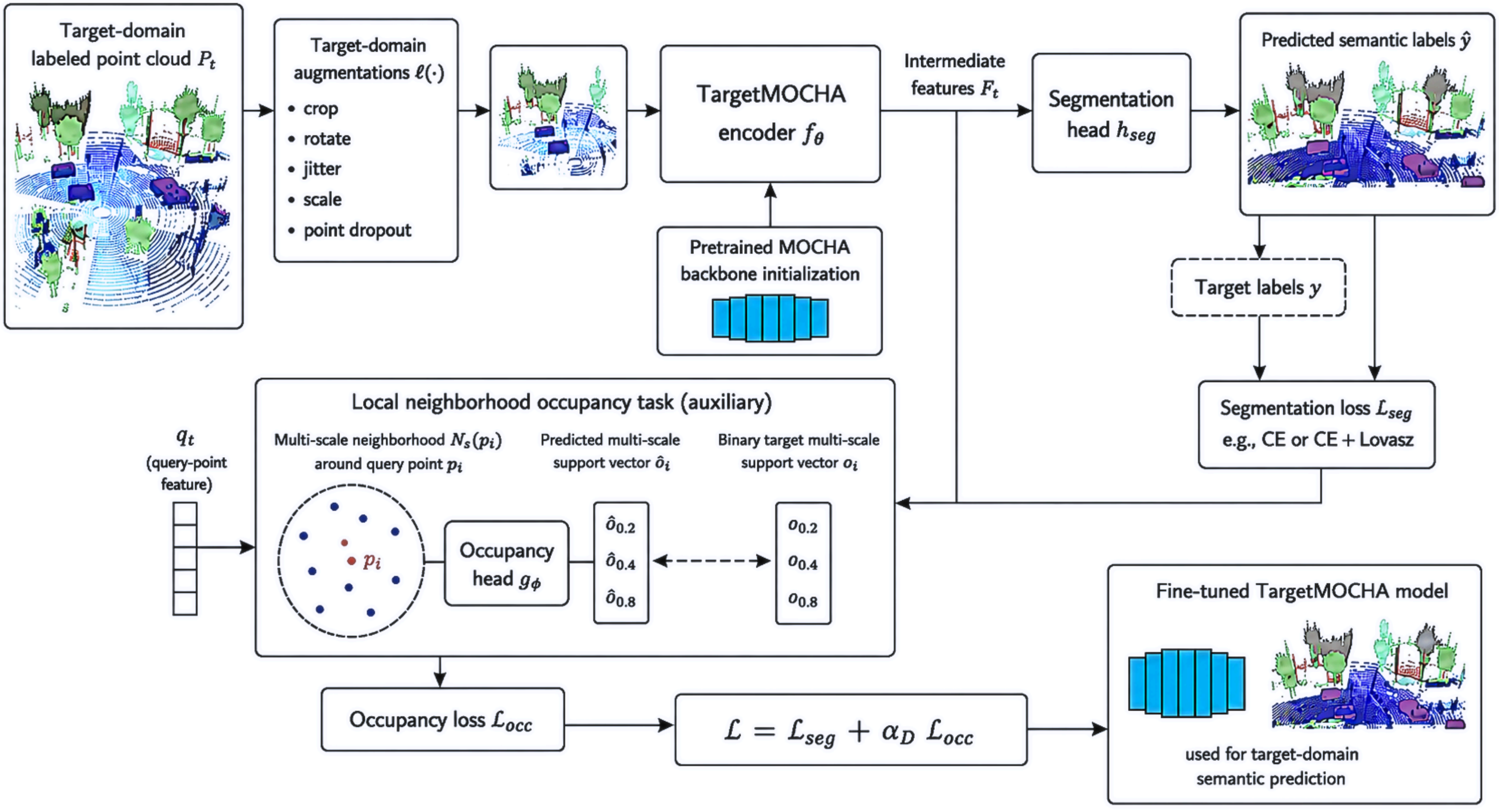

Target-domain labeled point cloud P_t
Target-domain augmentations $\ell(\cdot)$
• crop
• rotate
• jitter
• scale
• point dropout
TargetMOCHA encoder f_θ
Pretrained MOCHA backbone initialization
Intermediate features F_t
Segmentation head h_{seg}
Predicted semantic labels $\hat{y}$
Target labels y
Segmentation loss $\mathcal{L}_{seg}$
e.g., CE or CE + Lovasz
Local neighborhood occupancy task (auxiliary)
q_t
(query-point feature)
Multi-scale neighborhood $N_s(p_i)$ around query point p_i
Predicted multi-scale support vector $\hat{o}_i$
Binary target multi-scale support vector o_i
p_i
Occupancy head g_ϕ
$\hat{o}_{0.2}$
$\hat{o}_{0.4}$
$\hat{o}_{0.8}$
$o_{0.2}$
$o_{0.4}$
$o_{0.8}$
Occupancy loss $\mathcal{L}_{occ}$
$\mathcal{L} = \mathcal{L}_{seg} + \alpha_D \mathcal{L}_{occ}$
Fine-tuned TargetMOCHA model
used for target-domain semantic prediction